

figures-referenced

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Packages

```
library(tidyverse)
library(here)
library(janitor)
library(viridis)
library(wesanderson)
library(lme4)
library(ggplot2)
library(patchwork)
library(ggpmisc)
library(ggsignif)
```

Data

```
compiled_isotope <- read_csv(here("data", "compiled.csv"), show_col_types = FALSE) |>
  clean_names() |> # lowercase and underscores in column names
  rename(sample = sample_name, # renaming columns to be shorter
         delta_d = processed_delta_d,
         delta_d_std = processed_delta_d_stddev,
         delta_o = processed_delta_18o,
         delta_o_std = processed_delta_18o_stddev) |>
  mutate(date = mdy(date)) # telling r the date format

gradient_tc <- read_csv(here("data", "tc_gradient_data.csv"))

gradient_cc <- read_csv(here("data", "cc_gradient_data.csv"))

gradient_it <- read_csv(here("data", "it_gradient_data.csv"))
```

```

metadata <- read_csv(here("data", "metadata.csv"))

site_order <- c("tc_us", "tc_mid", "tc_ds", "tc_1", "tc_15", "tc_2", "tc_3")

trail_creek_isco <- compiled_isotope |>
  filter(site %in% c("utc_isco", "tc_isco"))

trail_creek <- compiled_isotope |>
  filter(site %in% site_order) |>
  mutate(site = factor(site, levels = site_order)) |>
  arrange(site)

coal_creek <- compiled_isotope |>
  filter(location %in% c("cc"))

italian_creek <- compiled_isotope |>
  filter(location == "it")

trail_creek_isco <- read_csv(here("data", "trail_creek_compiled_ISCO.csv")) |>
  clean_names() |> # lowercase and underscores in column names
  rename(sample = sample_name, # renaming columns to be shorter
    delta_d = processed_delta_d,
    delta_d_std = processed_delta_d_stddev,
    delta_o = processed_delta_18o,
    delta_o_std = processed_delta_18o_stddev) |>
  mutate(date = mdy(date)) # telling r the date format

utc_isco <- trail_creek_isco |>
  filter(site == "utc_isco")

tc_isco <- trail_creek_isco |>
  filter(site == "tc_isco")

# Downstream Trail Creek
tc_metrics <- read_csv(here("data", "TrailCreek_2026-07-23-15.csv"), show_col_types = FALSE)
  clean_names() |> # lowercase and underscores in column names
  rename(date_time = date_time_mdt, # renaming columns to be shorter
    dp = differential_pressure_k_pa,
    ap = absolute_pressure_k_pa,
    temp = temperature_c,
    water_level = water_level_m,

```

```

    bp = barometric_pressure_k_pa) |>
mutate(date_time_parsed = mdy_hms(date_time), # telling r the date format
       date = as_date(date_time_parsed)) |> # making a date column
group_by(date) |>
summarise(avg_dp = mean(dp, na.rm = TRUE), # averaging the data based on the day
          avg_ap = mean(ap, na.rm = TRUE),
          avg_temp = mean(temp, na.rm = TRUE),
          avg_water_level = mean(water_level, na.rm = TRUE),
          avg_bp = mean(bp, na.rm = TRUE),
          total_readings = n(), # the amount of data it used to average
          .groups = "drop") |>
mutate(site = "tc") # creating a new column of the site

# Upstream Trail Creek
utc_metrics <- read_csv(here("data", "UTC_2026-07-23-15.csv"), show_col_types = FALSE) |> # 1
  clean_names() |> # lowercase and underscores in column names
  rename(date_time = date_time_mdt, # renaming columns to be shorter
         dp = differential_pressure_k_pa,
         ap = absolute_pressure_k_pa,
         temp = temperature_c,
         water_level = water_level_m,
         bp = barometric_pressure_k_pa) |>
mutate(date_time_parsed = mdy_hms(date_time), # telling r the date format
       date = as_date(date_time_parsed)) |> # making a date column
group_by(date) |>
summarise(avg_dp = mean(dp, na.rm = TRUE), # averaging the data based on the day
          avg_ap = mean(ap, na.rm = TRUE),
          avg_temp = mean(temp, na.rm = TRUE),
          avg_water_level = mean(water_level, na.rm = TRUE),
          avg_bp = mean(bp, na.rm = TRUE),
          total_readings = n(), # the amount of data it used to average
          .groups = "drop") |>
mutate(site = "utc") # creating a new column of the site

tc_utc_metrics <- merge(tc_metrics, utc_metrics, all = TRUE) # merging the tc and utc data

isco_data <- merge(
  x = tc_utc_metrics,
  y = compiled_isotope,
  by.x = c("site", "date"), # columns in tc_utc_data
  by.y = c("location", "date") # matching columns in compiled_isotope
)

```

```
discharge_tc_isco <- read_csv(here("data", "discharge_tc_isco.csv"))
```

Figures

Figure 1: Time-Series of Line-Conditioned Excess (lc-excess) of ISCO's

```
ggplot(data = trail_creek_isco,
       aes(x = date,
           y = lc_excess,
           color = site,
           group = site)) +
  geom_hline(yintercept = 0,
            linetype = "dashed",
            color = "gray50",
            size = 0.8) +
  geom_point(size = 2.5,
            alpha = 0.8) +
  geom_smooth(method = "lm", se = TRUE, alpha = 0.15, linewidth = 1) +

  # adds the equation and R-squared text automatically for each date line
  stat_poly_eq(
    aes(label = paste(after_stat(eq.label), after_stat(rr.label), sep = "*\n", \n"*")),
    formula = y ~ x,
    parse = TRUE,
    label.x = "left", # positions the labels horizontally
    label.y = "top", # positions the labels vertically
    vstep = 0.05, # spacing between the text lines so they do not overlap
    size = 3.5) +
  scale_color_manual(
    values = c("tc_isco" = "dodgerblue", "utc_isco" = "orangered2"),
    labels = c("tc_isco" = "Trail Creek", "utc_isco" = "Upper Trail Creek")
  ) +

  labs(title = "Time-Series of Line-Conditioned Excess (lc-excess)",
       subtitle = "Values below 0 indicate potential evaporative enrichment",
       x = "Sampling Date",
       y = "lc-excess (\u2030)",
       color = "Site") +
  ylim(-8,5) +
  theme_bw() +
```

```
theme(plot.title = element_text(face = "bold", size = 14),
      panel.grid.minor = element_blank(),
      legend.position = "top")
```

Time-Series of Line-Conditioned Excess (lc-excess)

Values below 0 indicate potential evaporative enrichment

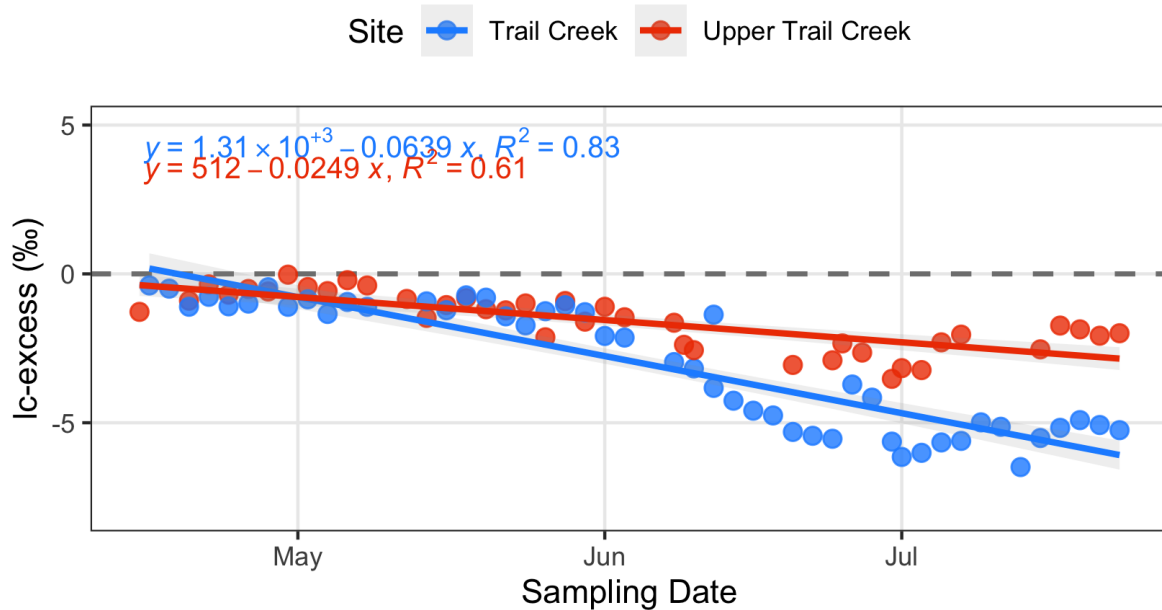


Figure 1.1: Sampling Site and lc-excess (Trail Creek)

```
plot1_1 <- ggplot(data = trail_creek, aes(x = site, y = lc_excess, color = factor(date), group = date)) +
  # reference line at 0
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray", linewidth = 0.8) +
  # adds a linear regression line for each date group
  geom_smooth(method = "lm", se = TRUE, linewidth = 1, alpha = 0.15) +
  # adds the equation and R-squared text automatically for each date line
  stat_poly_eq(
    aes(label = paste(after_stat(eq.label), after_stat(rr.label), sep = "*\n", \n")),
    formula = y ~ x,
    parse = TRUE,
    label.x = "left", # positions the labels horizontally
    label.y = "bottom", # positions the labels vertically
    vstep = 0.05, # spacing between the text lines so they do not overlap
    size = 3.0) +
```

```

# points for each site's value
geom_point(size = 2.0, alpha = 0.8) +
labs(
  title = "Trail Creek",
  x = "Sampling Site (Upstream to Downstream)",
  y = "lc-excess (\u2030)",
  color = "Sampling Date"
) +
scale_color_manual(values = c("2026-06-24" = "royalblue3", "2026-06-30" = "royalblue1", "2026-07-15" = "royalblue2"),
scale_x_discrete(labels = c("tc_us" = "TC-US", "tc_mid" = "TC-MID", "tc_ds" = "TC-DS", "bda_1" = "BDA-1", "bda_1.5" = "BDA-1.5", "bda_2" = "BDA-2", "bda_3" = "BDA-3")),
theme_bw() +
theme(
  plot.title = element_text(face = "bold", size = 14),
  panel.grid.minor = element_blank(),
  axis.text.x = element_text(angle = 45, hjust = 1),
  legend.position = "bottom"
)
plot1_1

```

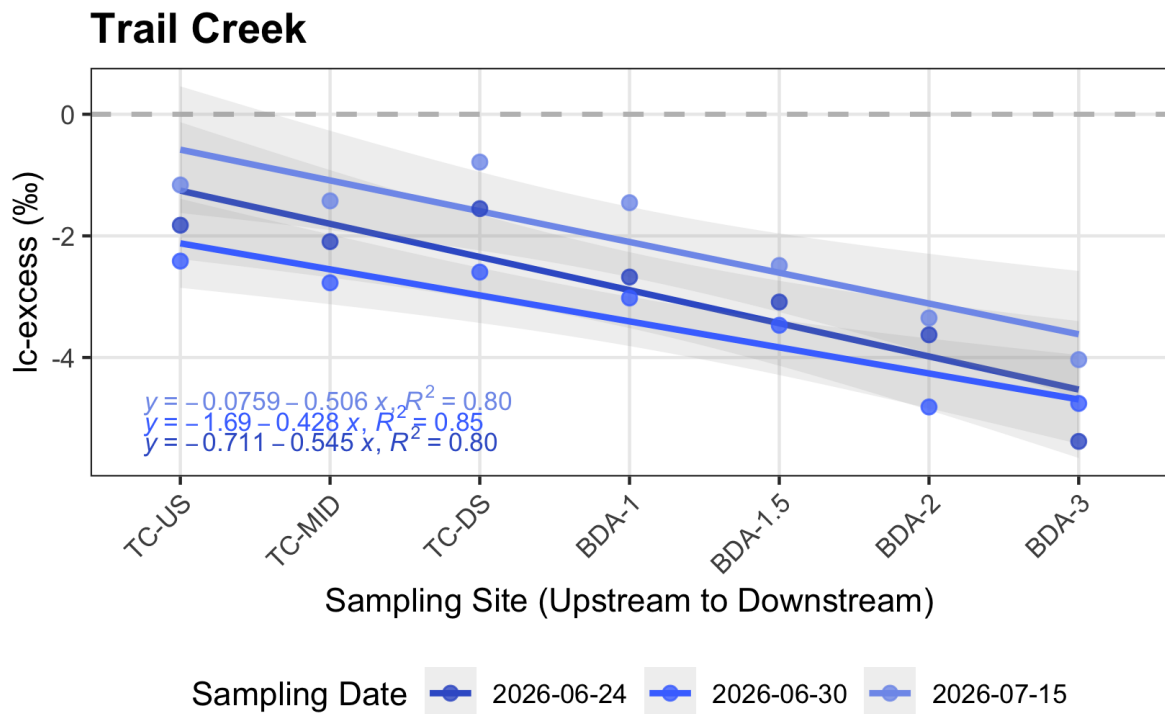


Figure 1.2: Sampling Site and lc-excess (Coal Creek)

```

plot1_2 <- ggplot(data = coal_creek, aes(x = site, y = lc_excess, color = factor(date), group = date)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray", linewidth = 0.8) +
  geom_smooth(method = "lm", formula = y ~ x, se = TRUE, linewidth = 1, alpha = 0.15) +
  stat_poly_eq(
    aes(label = paste(after_stat(eq.label), after_stat(rr.label), sep = "*\\", \\"*")),
    formula = y ~ x,
    parse = TRUE,
    label.x = "left",
    label.y = "bottom",
    vstep = 0.05,
    size = 3.0) +
  geom_point(size = 2.0, alpha = 0.8) +
  scale_color_manual(values = c("2026-06-23" = "royalblue3", "2026-07-07" = "royalblue1")) +
  scale_x_discrete(labels = c("coal_1" = "CC-1", "coal_2" = "CC-2", "coal_3" = "CC-3", "coal_4" = "CC-4")) +
  labs(
    title = "Coal Creek",
    x = "Sampling Site (Upstream to Downstream)",
    y = "lc-excess (\\u2030)",
    color = "Sampling Date"
  ) +
  theme_bw() +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    panel.grid.minor = element_blank(),
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "bottom"
  )
plot1_2

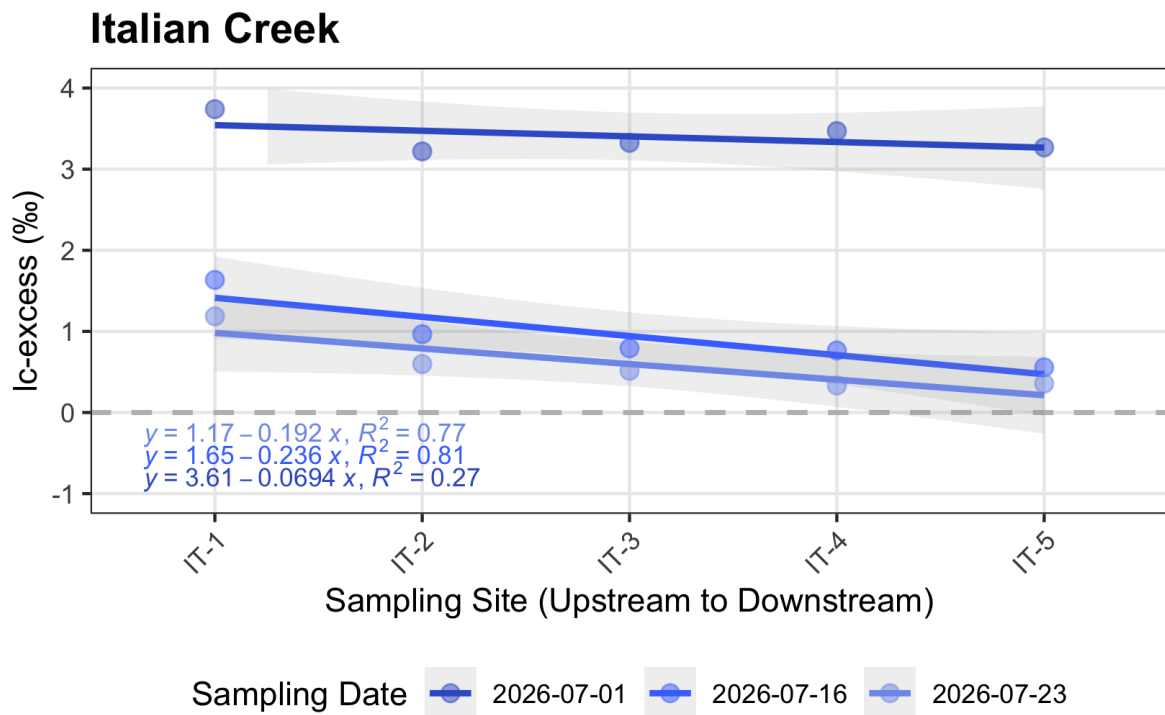
```



```

labs(
  title = "Italian Creek",
  x = "Sampling Site (Upstream to Downstream)",
  y = "lc-excess (\u2030)",
  color = "Sampling Date"
) +
theme_bw() +
theme(
  plot.title = element_text(face = "bold", size = 14),
  panel.grid.minor = element_blank(),
  axis.text.x = element_text(angle = 45, hjust = 1),
  legend.position = "bottom"
)
plot1_3

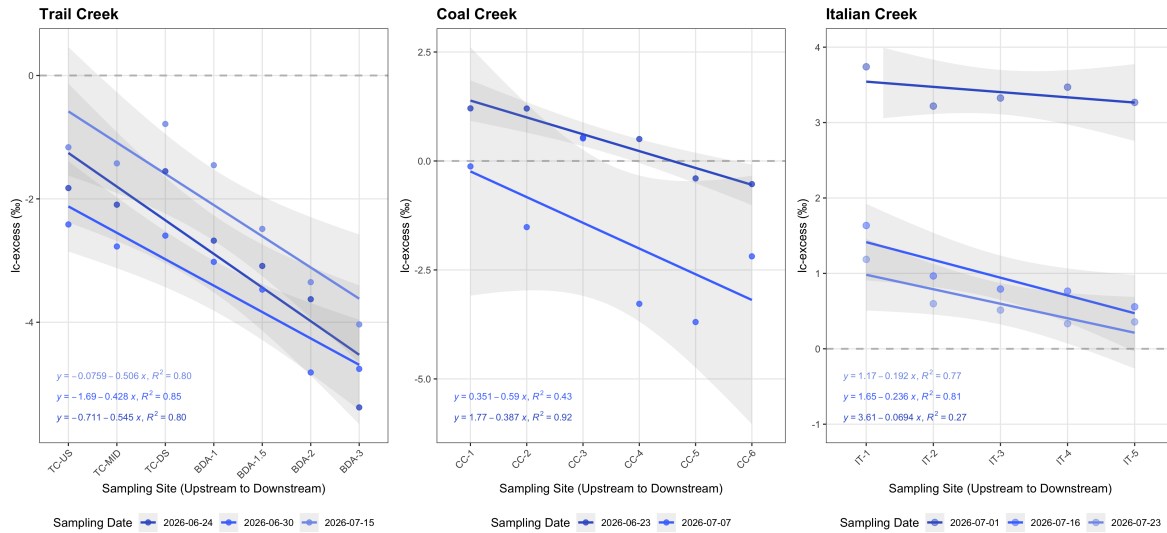
```



```

plot1_1 + plot1_2 + plot1_3

```



```
# Spatial variations in Δlc-excess/Δx across sampling dates
# Building a look up table of stream distances Trail Creek
distances <- tibble(site = c("TC-P3-US", "TC-P3-mid", "TC-P3-DS", "TC-BDA1",
                             "TC-BDA1.5", "TC-BDA2", "TC-BDA3"),
                    dist_m = c(57.6, 333, 654, 1160, 1620, 2240, 3150)) |>
  mutate(xmax = dist_m,
         xmin = lag(dist_m, default = 0)) # previous site's distance, 0 for first

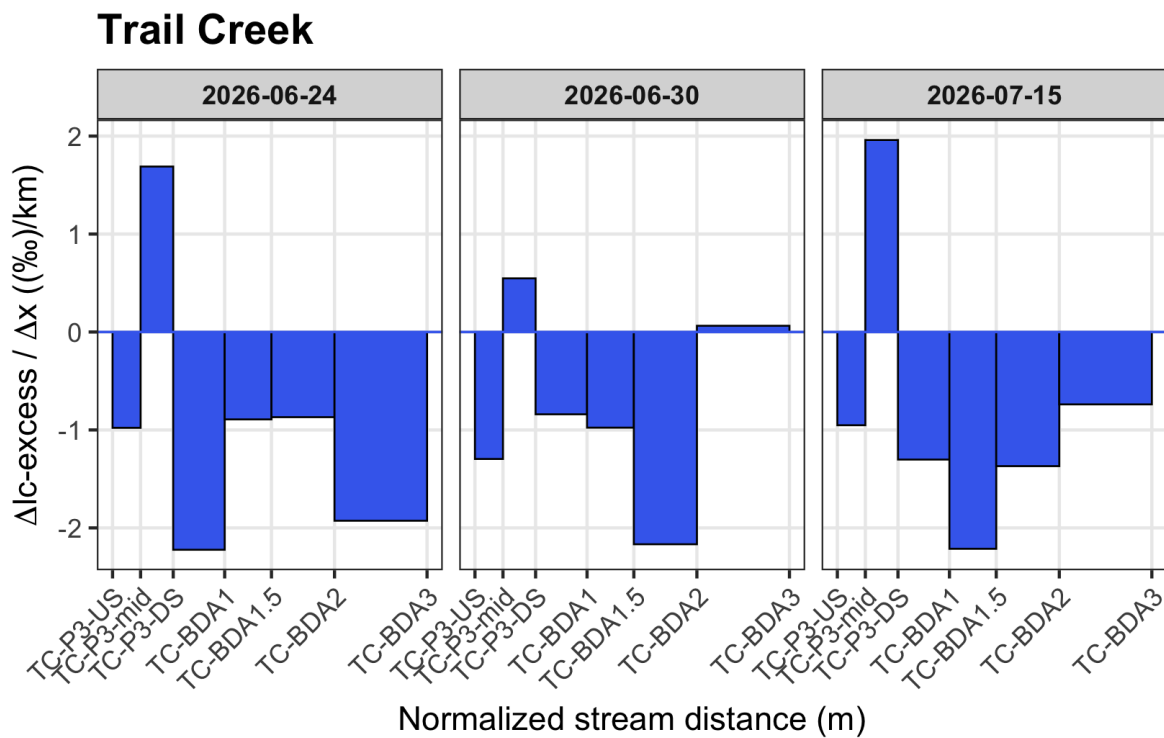
# Creating a filtered data frame for lc-excess
gradient_tc_clean <- gradient_tc |>
  select(site, normalized_stream_distance_m, `Δlc-excess/Δx_6-24`, `Δlc-excess/Δx_6-30`, `Δlc-excess/Δx_7-15`) |>
  pivot_longer( cols = c(`Δlc-excess/Δx_6-24`, `Δlc-excess/Δx_6-30`, `Δlc-excess/Δx_7-15`),
                names_to = "date",
                values_to = "gradient") |>
  mutate(date = case_when(date == "Δlc-excess/Δx_6-24" ~ "2026-06-24",
                          date == "Δlc-excess/Δx_6-30" ~ "2026-06-30",
                          date == "Δlc-excess/Δx_7-15" ~ "2026-07-15"),
         gradient = as.numeric(gradient)) |>
  left_join(distances %>% select(site, xmin, xmax), by = "site") |>
  mutate(site = factor(site, levels = distances$site)) |>
  filter(!is.na(gradient))

# Gradient plot
plot1 <- ggplot(gradient_tc_clean) +
  geom_rect(aes(xmin = xmin, xmax = xmax, ymin = 0, ymax = gradient),
            fill = "royalblue2", color = "black", linewidth = 0.3) +
```

```

geom_hline(yintercept = 0, color = "royalblue2", linewidth = 0.4) +
facet_wrap(~ date, nrow = 1) +
scale_x_continuous(
  breaks = distances$dist_m,
  labels = distances$site) +
labs(title = "Trail Creek",
  x = "Normalized stream distance (m)",
  y = expression(paste(Delta, "lc-excess / ", Delta, "x ((\u2030)/km)"))) +
theme_bw(base_size = 11) +
theme(plot.title = element_text(face = "bold", size = 14),
  panel.grid.minor = element_blank(),
  strip.text = element_text(face = "bold"),
  axis.text.x = element_text(angle = 45, hjust = 1))
plot1

```



```

# Buildings a look up table of stream distances Coal Creek
distances_cc <- tibble(
  site = c("CC1", "CC2", "CC4", "CC5", "CC6"),
  dist_m = c(0, 1496, 2006, 2478, 2885)) |>
mutate(xmax = dist_m,

```

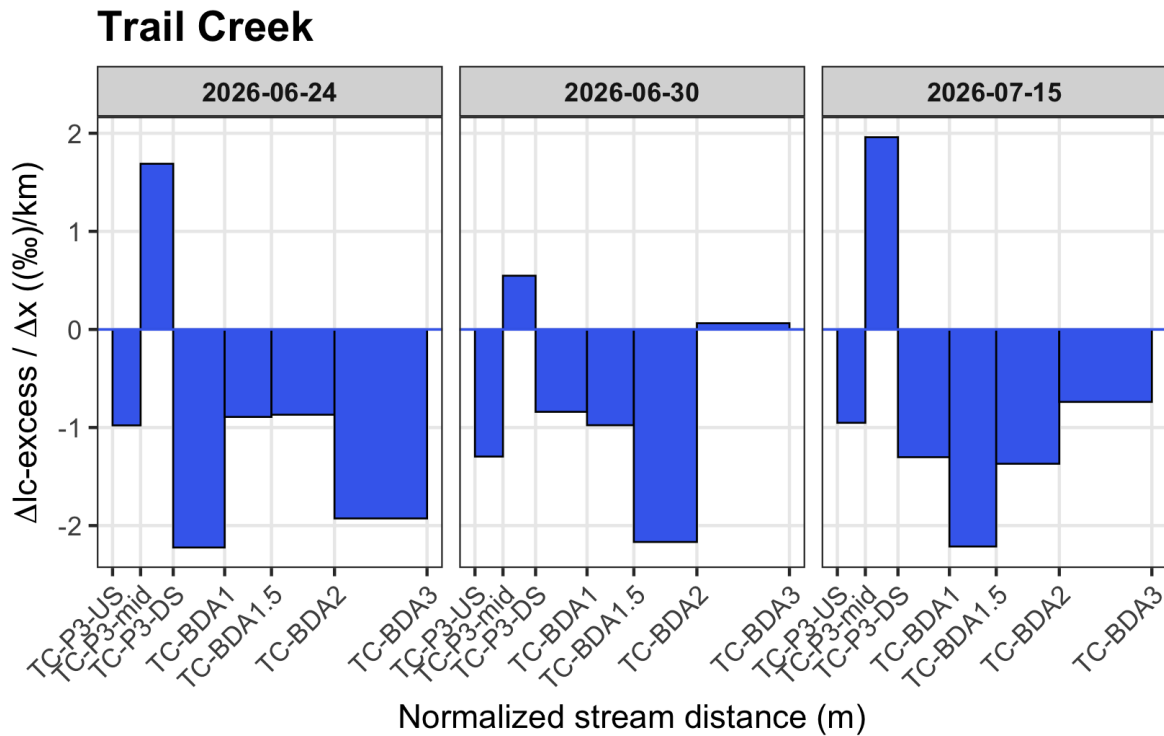
```

xmin = lag(dist_m, default = 0))

# Creating a filtered data frame for lc-excess
gradient_cc_clean <- gradient_cc |>
  select(site, normalized_stream_distance_m, `Δlc-excess/Δx_6-23`, `Δlc-excess/Δx_7-7`) |>
  pivot_longer(cols = c(`Δlc-excess/Δx_6-23`, `Δlc-excess/Δx_7-7`),
    names_to = "date",
    values_to = "gradient") |>
  mutate(date = case_when(
    date == "Δlc-excess/Δx_6-23" ~ "2026-06-23",
    date == "Δlc-excess/Δx_7-7" ~ "2026-07-07"),
    gradient = as.numeric(gradient)) |>
  left_join(distances_cc |> select(site, xmin, xmax),
    by = "site") |>
  mutate(site = factor(site, levels = distances_cc$site)) |>
  filter(!is.na(gradient))

```

plot1



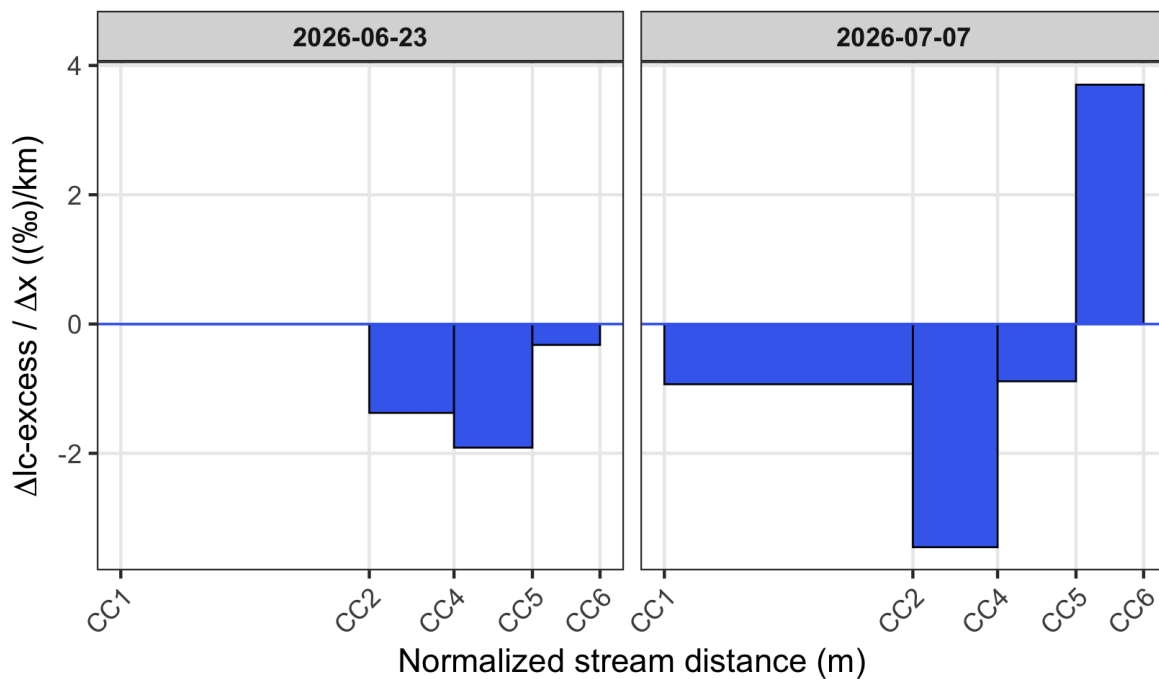
```

# Gradient plot
plot2 <- ggplot(gradient_cc_clean) +
  geom_rect(aes(xmin = xmin, xmax = xmax, ymin = 0, ymax = gradient),
            fill = "royalblue2", color = "black", linewidth = 0.3) +
  geom_hline(yintercept = 0, color = "royalblue2", linewidth = 0.4) +
  facet_wrap(~date, nrow = 1) +
  scale_x_continuous(
    breaks = distances_cc$dist_m,
    labels = distances_cc$site) +
  labs(
    title = "Coal Creek",
    x = "Normalized stream distance (m)",
    y = expression(paste(Delta, "lc-excess / ", Delta, "x ((\u2030)/km)"))) +
  theme_bw(base_size = 11) +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    panel.grid.minor = element_blank(),
    strip.text = element_text(face = "bold"),
    axis.text.x = element_text(angle = 45, hjust = 1))

plot2

```

Coal Creek



```
library(tidyverse)

# 1. Build the look-up table for Italian Creek
distances_it <- tibble(
  site = c("IT1", "IT2", "IT3", "IT4", "IT5"),
  dist_m = c(0, 793.37, 1665.83, 2385.13, 2963.62) # Keeps a clean width layout like plot1
) |>
mutate(
  xmax = dist_m,
  xmin = lag(dist_m, default = 0)
)

# 2. Creating a filtered data frame for Italian Creek (Fixed Column Names)
gradient_it_clean <- gradient_it |>
select(site, normalized_stream_distance_m, `d_lc/d_x_7_1`, `d_lc/d_x_7_16`, `d_lc/d_x_7_23`)
pivot_longer(
  cols = c(`d_lc/d_x_7_1`, `d_lc/d_x_7_16`, `d_lc/d_x_7_23`),
  names_to = "date",
  values_to = "gradient"
) |>
```

```

mutate(
  date = case_when(
    date == "d_lc/d_x_7_1" ~ "2026-07-01",
    date == "d_lc/d_x_7_16" ~ "2026-07-16",
    date == "d_lc/d_x_7_23" ~ "2026-07-23"
  ),
  gradient = as.numeric(gradient)
) |>
left_join(distances_it |> select(site, xmin, xmax),
          by = "site") |>
mutate(site = factor(site, levels = distances_it$site)) |> # FIXED: Uses distances_it inst
filter(!is.na(gradient))

# 3. Gradient plot for Italian Creek
plot2_5 <- ggplot(gradient_it_clean) +
  geom_rect(
    aes(xmin = xmin, xmax = xmax, ymin = 0, ymax = gradient),
    fill = "royalblue2", color = "black", linewidth = 0.3
  ) +
  geom_hline(yintercept = 0, color = "royalblue2", linewidth = 0.4) +
  facet_wrap(~ date, nrow = 1) +
  scale_x_continuous(
    breaks = distances_it$dist_m,
    labels = distances_it$site
  ) +
  labs(
    title = "Italian Creek",
    x = "Normalized stream distance (m)",
    y = expression(paste(Delta, "lc-excess / ", Delta, "x ((\u2030)/km)"))
  ) +
  theme_bw(base_size = 11) +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    panel.grid.minor = element_blank(),
    strip.text = element_text(face = "bold"),
    axis.text.x = element_text(angle = 45, hjust = 1)
  )

plot2_5

```

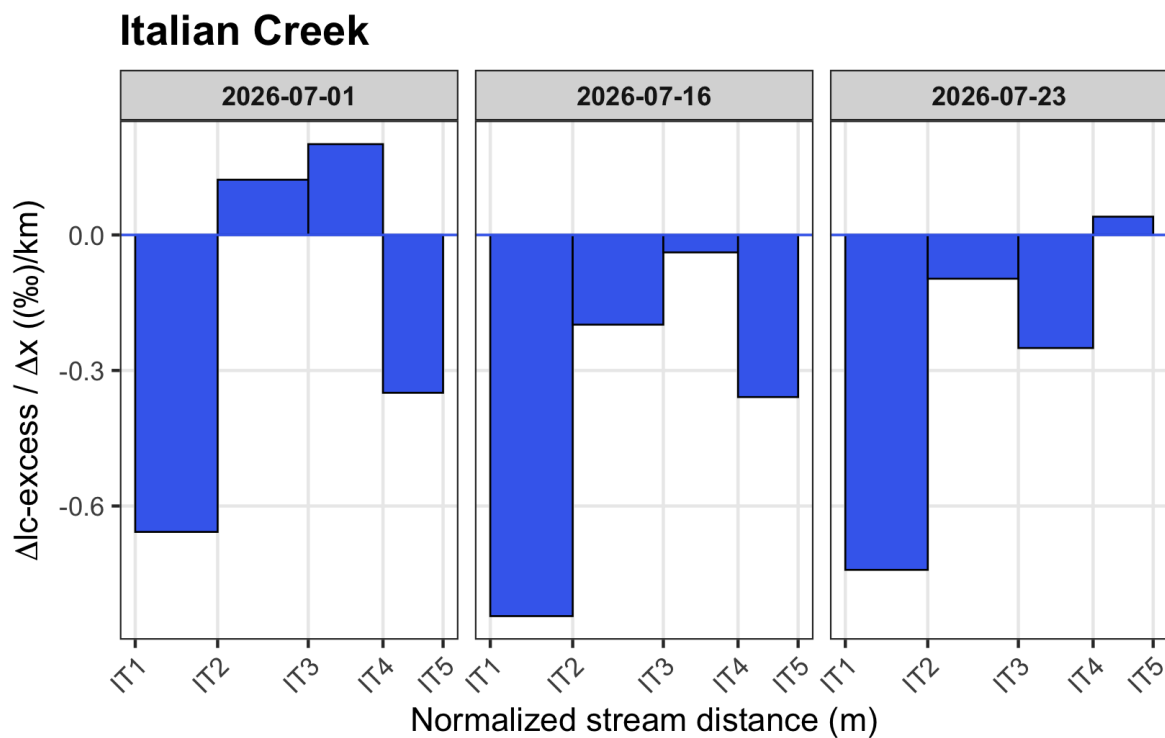


Figure 2:

```
combined_stacked <- plot1 + plot2 + plot2_5
combined_stacked # print the graph
```

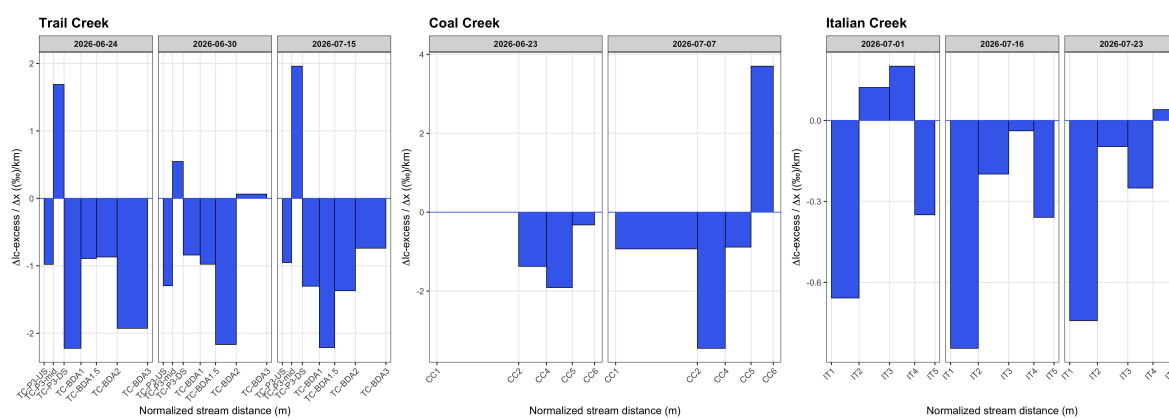
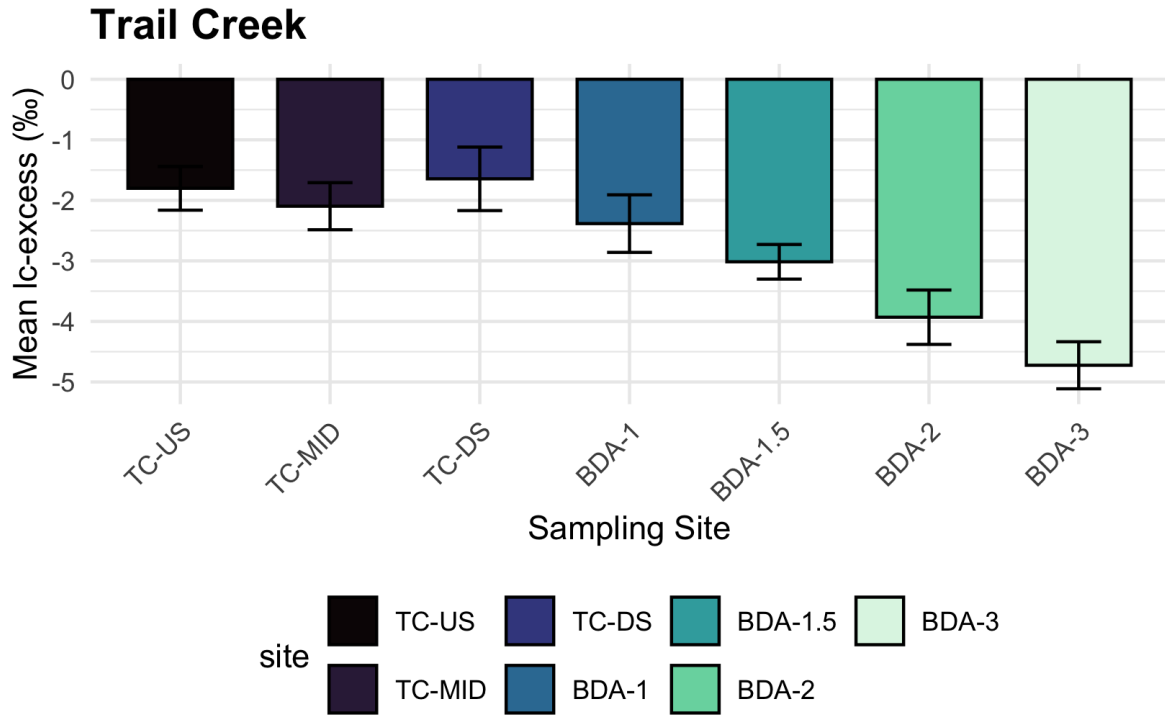


Figure 3

```
# averaging the lc-excess and including standard error
delta_summary <- trail_creek |>
  mutate(site = factor(site, levels = site_order)) |>
  group_by(site) |>
  summarise(
    mean_lc = mean(lc_excess, na.rm = TRUE),
    se_lc = sd(lc_excess, na.rm = TRUE) / sqrt(n()),
    .groups = "drop")

# creating a plot to show the mean lc-excess of each site (coal creek)
plot3 <- ggplot(delta_summary, aes(x = site, y = mean_lc, fill = site)) +
  geom_col(color = "black", width = 0.7) +
  geom_errorbar(aes(ymin = mean_lc - se_lc, ymax = mean_lc + se_lc),
    width = 0.3,
    color = "black") +
  scale_x_discrete(labels = c("tc_1" = "BDA-1", "tc_15" = "BDA-1.5",
    "tc_2" = "BDA-2", "tc_3" = "BDA-3",
    "tc_ds" = "TC-DS", "tc_mid" = "TC-MID",
    "tc_us" = "TC-US")) +
  scale_fill_viridis_d(option = "G", labels = c("tc_1" = "BDA-1", "tc_15" = "BDA-1.5", "tc_2" =
    "tc_3" = "BDA-3", "tc_ds" = "TC-DS", "tc_mid" = "TC-MID", "tc_us" = "TC-US")) +
  theme_minimal() +
  labs(title = "Trail Creek",
    x = "Sampling Site",
    y = "Mean lc-excess (\u2030)") +
  theme(plot.title = element_text(face = "bold", size = 14),
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "bottom")

plot3
```

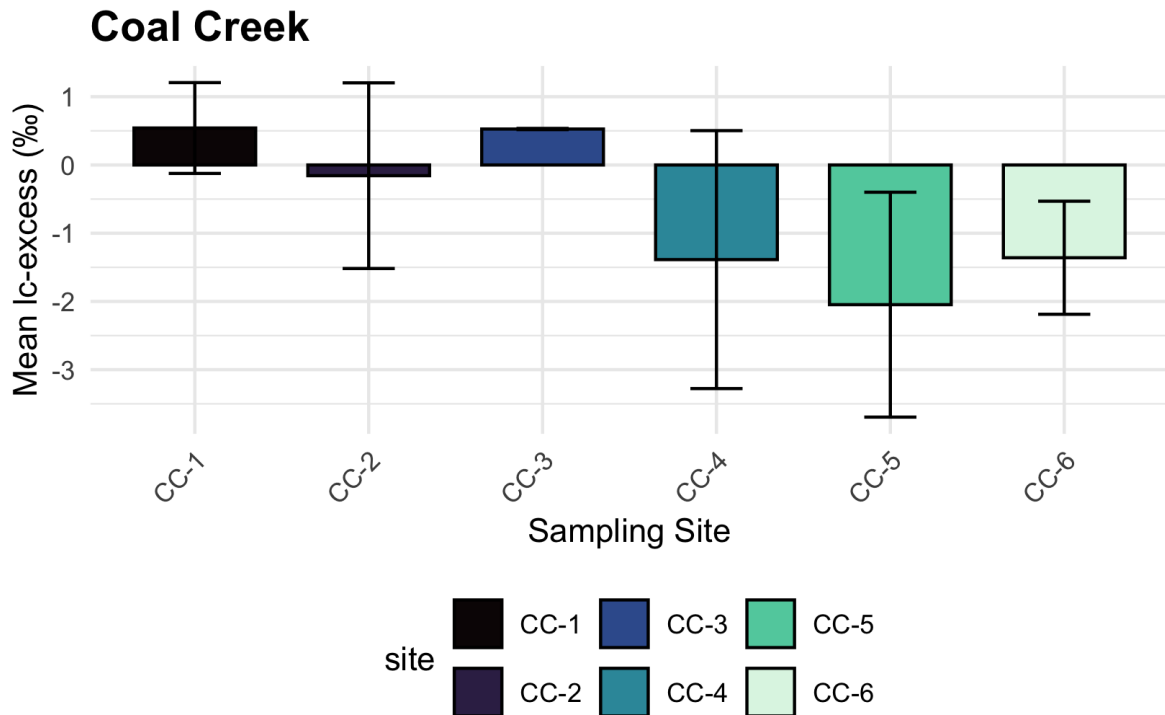


```
# averaging the lc-excess and including standard error
delta_sum_cc <- coal_creek |>
  group_by(site) |>
  summarise(
    mean_lc = mean(lc_excess, na.rm = TRUE),
    se_lc = sd(lc_excess, na.rm = TRUE) / sqrt(n()),
    .groups = "drop")

# creating a plot to show the mean lc-excess of each site (coal creek)
plot4 <- ggplot(delta_sum_cc, aes(x = site, y = mean_lc, fill = site)) +
  geom_col(color = "black", width = 0.7) +
  geom_errorbar(aes(ymin = mean_lc - se_lc, ymax = mean_lc + se_lc),
    width = 0.3, color = "black") +
  scale_fill_viridis_d(option = "G", labels = c("coal_1" = "CC-1", "coal_2" = "CC-2", "coal_3" = "CC-3", "coal_4" = "CC-4")) +
  scale_x_discrete(labels = c("coal_1" = "CC-1", "coal_2" = "CC-2", "coal_3" = "CC-3", "coal_4" = "CC-4")) +
  theme_minimal() +
  labs(title = "Coal Creek",
    x = "Sampling Site",
    y = "Mean lc-excess (‰)") +
  theme(plot.title = element_text(face = "bold", size = 14),
    axis.text.x = element_text(angle = 45, hjust = 1),
```

```
legend.position = "bottom")
```

plot4



```
# averaging the lc-excess and including standard error
```

```
delta_summary_it <- italian_creek |>
```

```
  group_by(site) |>
```

```
  summarise(
```

```
    mean_lc = mean(lc_excess, na.rm = TRUE),
```

```
    se_lc = sd(lc_excess, na.rm = TRUE) / sqrt(n()),
```

```
    .groups = "drop")
```

```
# creating a plot to show the mean lc-excess of each site (italian creek)
```

```
plot5 <- ggplot(delta_summary_it, aes(x = site, y = mean_lc, fill = site)) +
```

```
  geom_col(color = "black", width = 0.7) +
```

```
  geom_errorbar(aes(ymin = mean_lc - se_lc, ymax = mean_lc + se_lc),
```

```
    width = 0.3, color = "black") +
```

```
  scale_fill_viridis_d(option = "G", labels = c("italian_1" = "IT-1", "italian_2" = "IT-2", "italian_3" = "IT-3")) +
```

```
  scale_x_discrete(labels = c("italian_1" = "IT-1", "italian_2" = "IT-2", "italian_3" = "IT-3")) +
```

```
  theme_minimal() +
```

```
labs(title = "Italian Creek",
     x = "Sampling Site",
     y = "Mean lc-excess (‰)") +
theme(plot.title = element_text(face = "bold", size = 14),
     axis.text.x = element_text(angle = 45, hjust = 1),
     legend.position = "bottom")
```

plot5

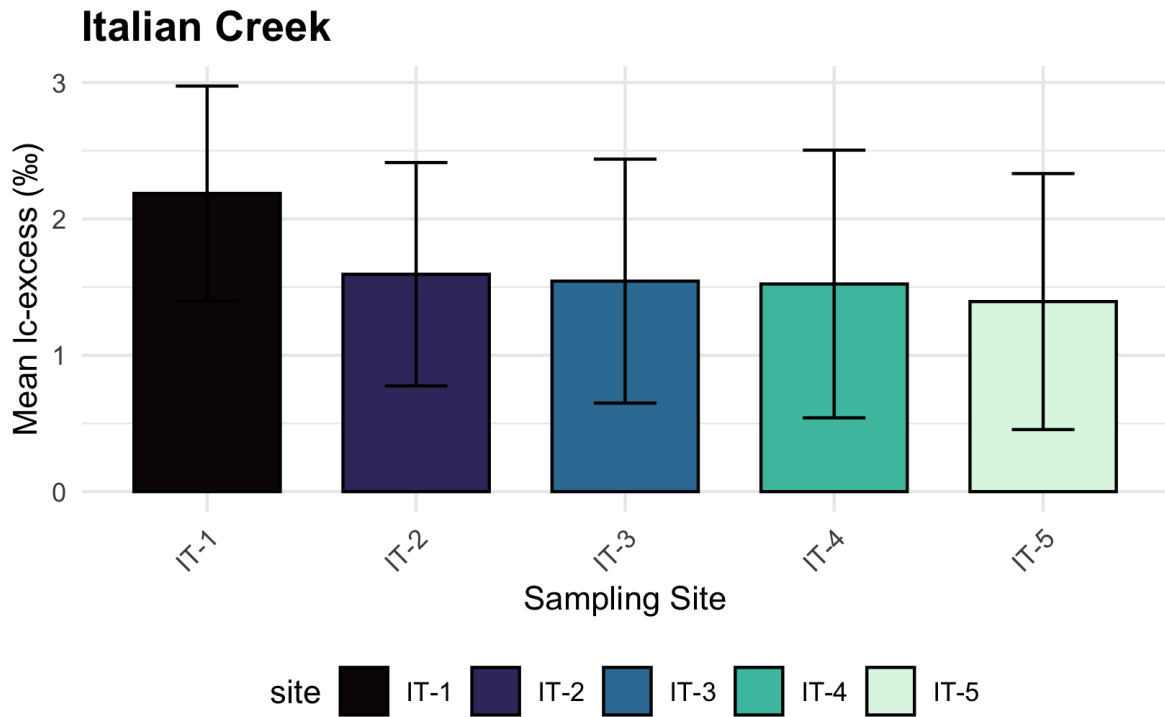


Figure 3:

```
combined_stacked_2 <- plot3 + plot4 + plot5

combined_stacked_2 # print the graph
```

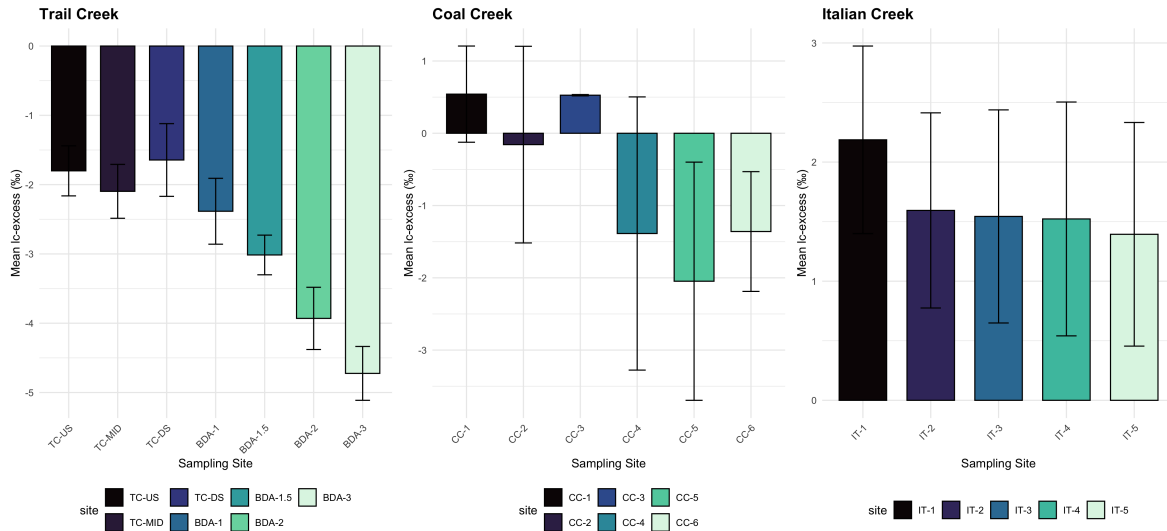
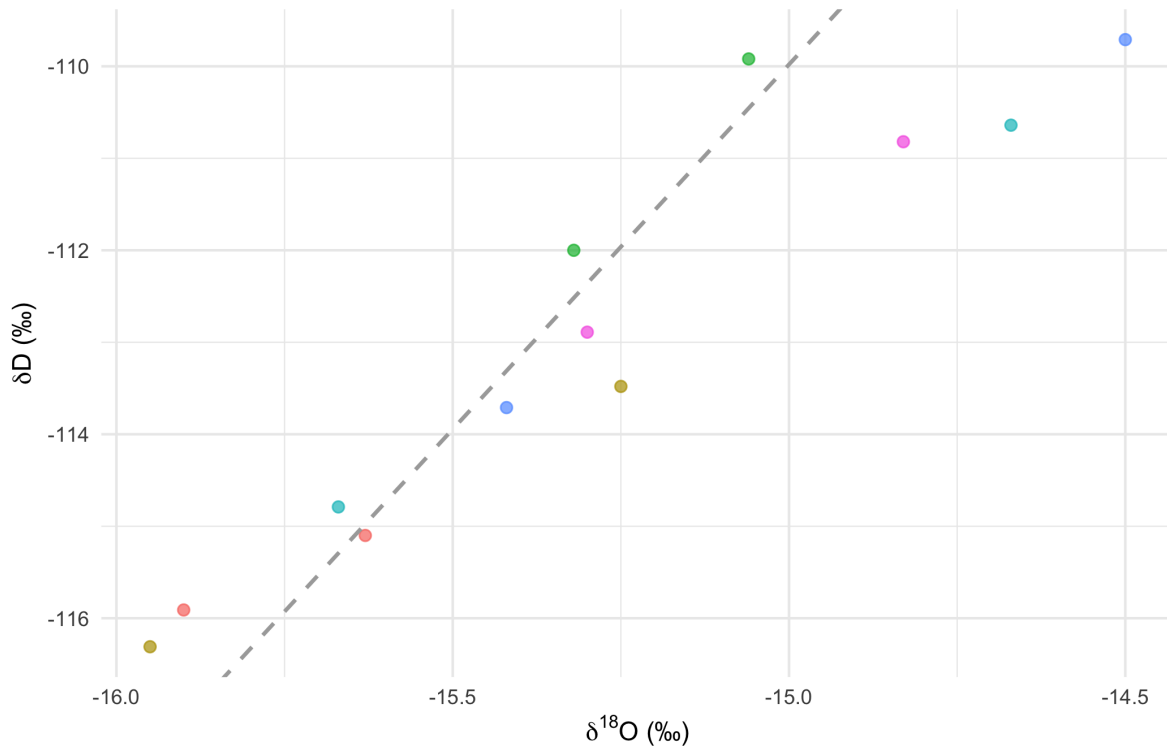


Figure 4:

```
# Coal Creek LMWL calculated from OPIC  $y = 7.93 x + 8.97$ 
ggplot(coal_creek, aes(x = delta_o, y = delta_d, color = site)) +
  geom_point(alpha = 0.7, size = 2) +
  geom_abline(intercept = 8.97, slope = 7.93, linetype = "dashed", color = "darkgray", linewidth = 1) +
  labs(
    title = "Dual-Isotope Plot:  $\delta^{18}O$  vs  $\delta^2H$ ",
    subtitle = "Dashed line = LMWL",
    x = expression(paste(delta^{18}, "O (\u2030)")),
    y = expression(paste(delta, "D (\u2030)")),
    color = "Site"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14), legend.position = "none")
```

Dual-Isotope Plot: $\delta^{18}\text{O}$ vs δD

Dashed line = LMWL

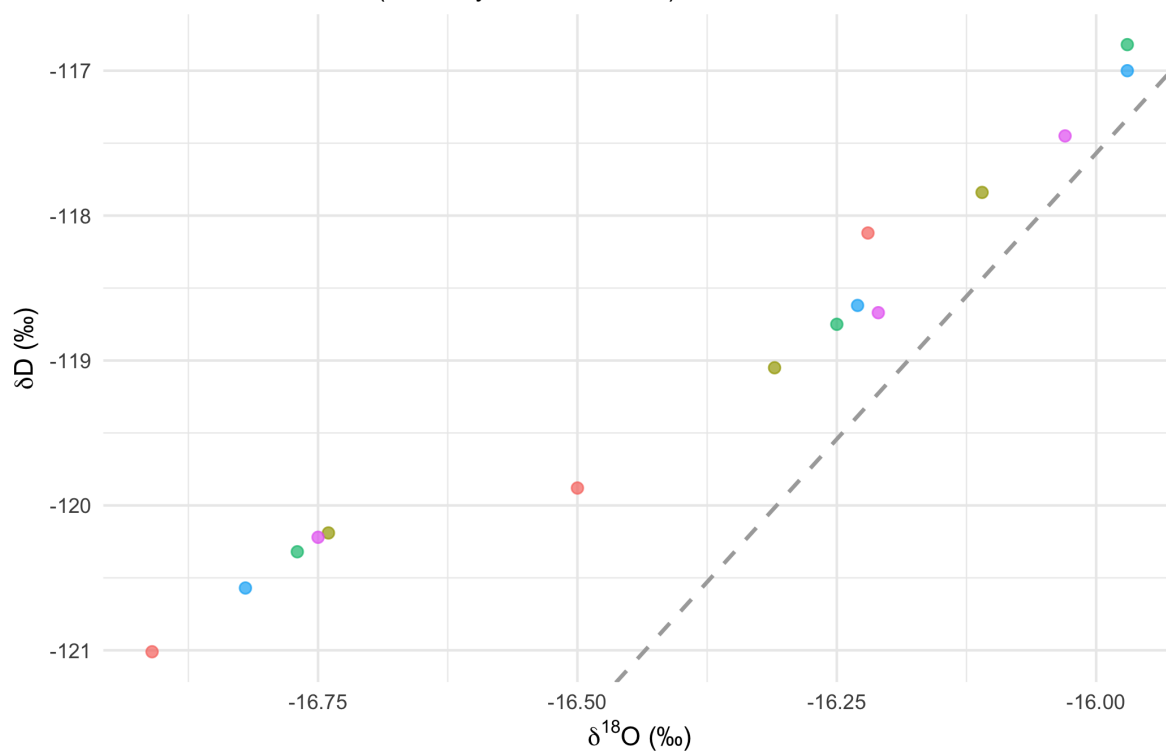


```
# Italian Creek LMWL calculated from OPIC  $y = 7.89 x + 8.67$ 

ggplot(italian_creek, aes(x = delta_o, y = delta_d, color = site)) +
  geom_point(alpha = 0.7, size = 2) +
  geom_abline(intercept = 8.67, slope = 7.89, linetype = "dashed", color = "darkgray", linewidth = 1) +
  labs(
    title = "Dual-Isotope Plot:  $\delta^{18}\text{O}$  vs  $\delta\text{D}$ ",
    subtitle = "Local Mean Water Line (LMWL:  $y = 7.89 x + 8.67$ )",
    x = expression(paste(delta^{18}, "O (‰)")),
    y = expression(paste(delta, "D (‰)")),
    color = "Site"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14), legend.position = "none")
```

Dual-Isotope Plot: $\delta^{18}\text{O}$ vs δD

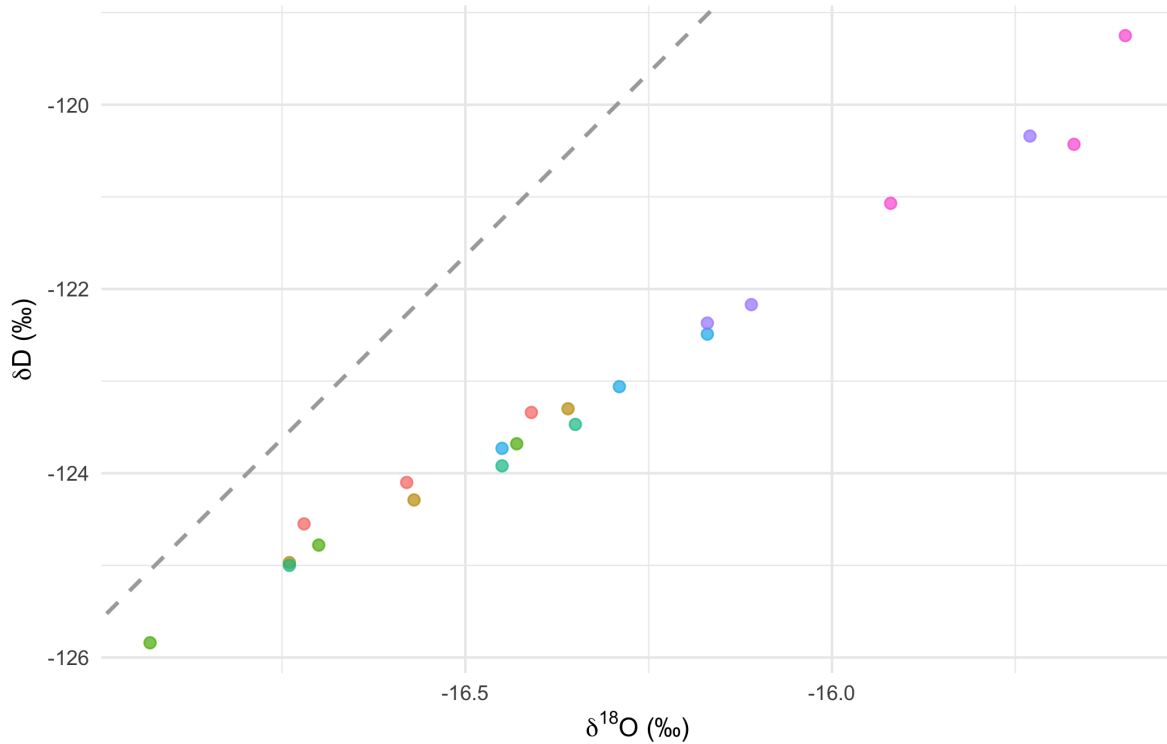
Local Mean Water Line (LMWL: $y = 7.89x + 8.67$)



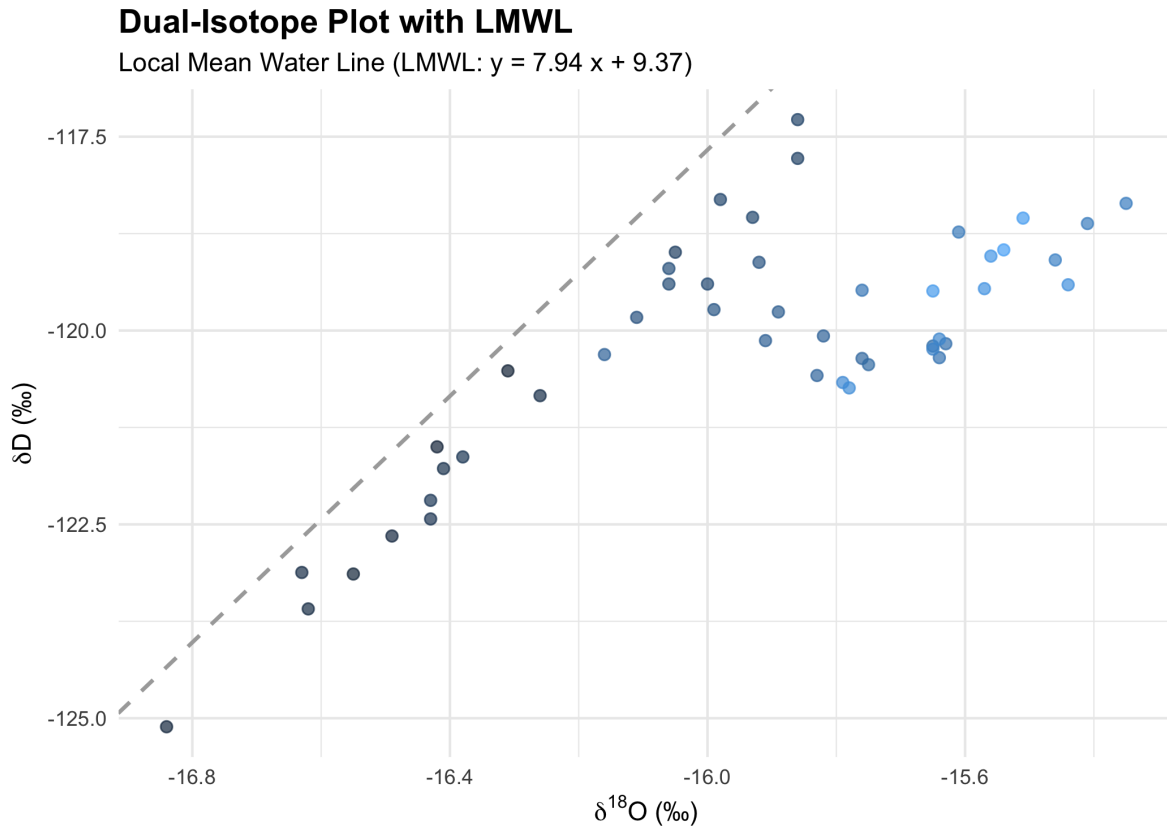
```
# Trail Creek River Reach LMWL calculated from OPIC  $y = 7.94x + 9.37$ 
ggplot(trail_creek, aes(x = delta_o, y = delta_d, color = site)) +
  geom_point(alpha = 0.7, size = 2) +
  geom_abline(intercept = 9.37, slope = 7.94, linetype = "dashed", color = "darkgray", linewidth = 1) +
  labs(
    title = "Dual-Isotope Plot with LMWL",
    subtitle = "Local Mean Water Line (LMWL:  $y = 7.94x + 9.37$ )",
    x = expression(paste(delta^{18}, "O (‰)")),
    y = expression(paste(delta, "D (‰)")),
    color = "Site"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14), legend.position = "none")
```

Dual-Isotope Plot with LMWL

Local Mean Water Line (LMWL: $y = 7.94 x + 9.37$)



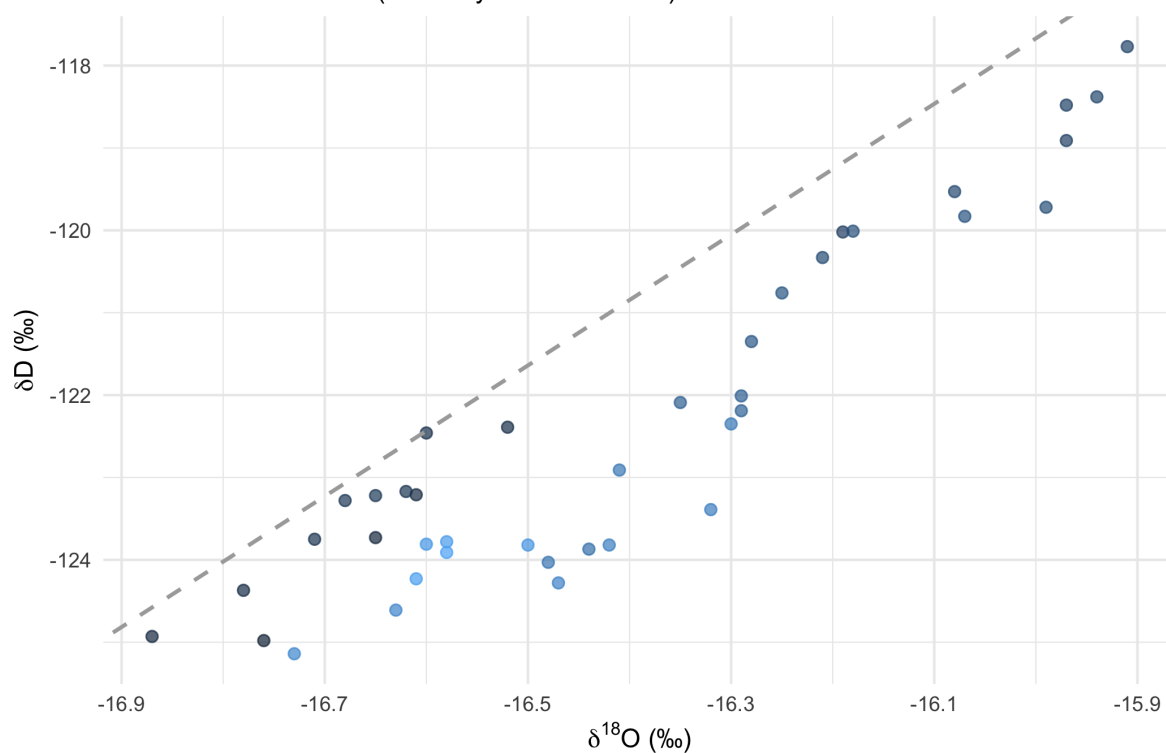
```
# TC ISCO LMWL calculated from OPIC  $y = 7.94 x + 9.37$ 
ggplot(tc_isco, aes(x = delta_o, y = delta_d, color = date)) +
  geom_point(alpha = 0.7, size = 2) +
  geom_abline(intercept = 9.37, slope = 7.94, linetype = "dashed", color = "darkgray", linewidth = 2) +
  labs(
    title = "Dual-Isotope Plot with LMWL",
    subtitle = "Local Mean Water Line (LMWL:  $y = 7.94 x + 9.37$ )",
    x = expression(paste(delta^{18}, "O (‰)")),
    y = expression(paste(delta, "D (‰)")),
    color = "Site"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14), legend.position = "none")
```

```
# UTC ISCO LMWL calculated from OPIC  $y = 7.94x + 9.37$ 
ggplot(utc_isco, aes(x = delta_o, y = delta_d, color = date)) +
  geom_point(alpha = 0.7, size = 2) +
  geom_abline(intercept = 9.37, slope = 7.94, linetype = "dashed", color = "darkgray", linewidth = 1) +
  labs(
    title = "Dual-Isotope Plot with LMWL",
    subtitle = "Local Mean Water Line (LMWL:  $y = 7.94x + 9.37$ )",
    x = expression(paste(delta^{18}, "O (‰)")),
    y = expression(paste(delta, "D (‰)")),
    color = "Site"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14), legend.position = "none")
```

Dual-Isotope Plot with LMWL

Local Mean Water Line (LMWL: $y = 7.94x + 9.37$)



Discharge

```
model_discharge_tc <- lm(lc_excess ~ discharge, data = discharge_tc_isco)
summary(model_discharge_tc)
```

Call:

```
lm(formula = lc_excess ~ discharge, data = discharge_tc_isco)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.13810	-0.33678	-0.07501	0.29166	1.00884

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-5.986	0.145	-41.28	<2e-16 ***
discharge	42.209	1.719	24.55	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5488 on 44 degrees of freedom

Multiple R-squared: 0.932, Adjusted R-squared: 0.9304

F-statistic: 602.9 on 1 and 44 DF, p-value: < 2.2e-16

```
ggplot(discharge_tc_isco, aes(x = discharge, y = lc_excess)) +  
  geom_point(size = 2.5, alpha = 0.7, color = "#2a78d6") +  
  geom_smooth(method = "lm", formula = y ~ x, se = TRUE, color = "black", linetype = "dashed") +  
  stat_poly_eq(  
    use_label(c("eq", "R2", "P")),  
    formula = y ~ x,  
    label.x = "left", label.y = "top",  
    size = 3.5  
  ) +  
  labs(title = "TC-ISCO: Discharge vs. lc-excess",  
       x = expression(paste("Discharge (cfs)")),  
       y = "lc-excess (\u2030)") +  
  theme_bw()
```

